A Reinforced Neighborhood Search Method Combined With Genetic Algorithm for Multi-Objective Multi-Robot Transportation System

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***Abstract*— With the rapid advancement of artificial intelligence, autonomous multi-robot systems have been successfully applied to various domains. Therefore, developing intelligent routing and scheduling systems to efficiently coordinate multi-robot movements in transportation networks emerges as a critical challenge. To address this issue, this study constructs an optimization model for cooperative robot operations, aiming to minimize total energy consumption and the completion time of most time-consuming robot. These objectives contain conflicts, thus requiring a multi-objective optimization approach to resolve them. We propose a reinforced neighborhood search method combined with genetic algorithm (RNSGA), which combines single solution search ideas and population-based techniques. RNSGA consists of two crucial steps: route construction to determine the composition and visiting sequence of task points within each route, as well as route allocation to assign routes to individual robots. The route construction phase incorporates several key components, including solution initialization, route balance mechanism, proximity-based optimization mechanism, and intro-route sequence adjustment method. For the route allocation phase, a population-based allocation mechanism is employed to determine the optimal assignment of routes. Comprehensive experiments on 24 classic transportation test instances demonstrate that RNSGA significantly outperforms six state-of-the-art algorithms.**

***Index Terms*— Multi-robot systems, multi-objective optimization, routing and scheduling, neighborhood search, genetic algorithm.**

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I. Introduction

THE integration of robotics and artificial intelligence has revolutionized traditional industries, leading to immense improvements in production and logistics efficiency [1]. From urban delivery to factory distribution, automatic robots are becoming increasingly vital in addressing complex operational challenges [2], [3], [4]. While various robots have been developed to reduce manual intervention in both operational and transportation tasks [5], single-robot systems prove inefficient when handling all tasks in large-scale scenarios [6]. Consequently, the trend has shifted toward multi-robot collaborative systems [7].

Despite the rapid development of autonomous technologies, efficient multi-robot routing and scheduling (MRRS) in complex transportation networks remains challenging [8]. In urban logistics scenarios, robots need to handle pickup and delivery operations at multiple locations while considering their payload capacity constraints. Due to the large number of tasks and limited robot capacity, operations must be efficiently distributed among multiple robots, with each robot typically requiring multiple trips to complete its assigned tasks.

The MRRS problem encompasses two key steps: the determination of multiple optimal routes, including both composition of each route and task points visiting sequence, and the efficient scheduling of these routes among available robots. From a broader perspective, this MRRS problem can be conceptualized as an advanced variant of the multi-trip vehicle routing problem (MTVRP) [9], which inherently belongs to the class of NP-hard problems. Additionally, practical transportation characteristics must be incorporated, particularly considering system congestion and the effects of varying payload on robot energy consumption and travel

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time. While traditional MTVRP typically focuses on energy consumption [10], makespan optimization serves as equally a crucial metric in practical logistics scenarios. Consequently, the complexity of this study significantly surpasses that of conventional MTVRP due to its multi-objective nature and real-world constraints.

Given the computational complexity of exact algorithms for such problems [11], we propose a reinforced neighborhood search method combined with genetic algorithm (RNSGA). This hybrid approach leverages the complementary strengths of single-solution search strategies and population-based techniques, enabling effective exploration of the complex solution space while maintaining solution diversity. The main contributions of this study are as follows:

- A mathematical model incorporating practical operational constraints is proposed to address the multi-robot routing and scheduling problem.
- A reinforced neighborhood search method combined with genetic algorithm is proposed to optimize the route construction and scheduling for multiple robots.
- Extensive comparative experiments demonstrate the effectiveness and superiority of RNSGA on 24 classic transportation test instances.

The structure of this paper is as follows: Section II reviews the relevant literature. Section III provides a problem description. Section IV introduces the proposed RNSGA algorithm. Section V designs experiments and analyzes results. Finally, Section VI summarizes the paper and outlines future works.

II. Literature Review

The MRRS problem investigated in this study extends the MTVRP by incorporating certain characteristics of multi-robot task allocation (MRTA) [12]. MTVRP emphasizes the routing aspects where vehicles make multiple trips subject to capacity constraints [13], while MRTA focuses on assigning tasks to multiple robots efficiently [14]. Our problem inherits characteristics from both: the multi-trip nature from MTVRP and the route allocation complexity from MRTA, while introducing additional transportation-specific constraints. This section reviews related literature from three perspectives: MTVRP approaches, MRTA approaches, and MRRS approaches.

A. MTVRP Approaches

The MTVRP typically emerges in urban logistics and distribution systems, where vehicles conduct multiple trips to serve customers within confined geographical areas. Initial research in this domain can be traced to Fleischmann et al. [15], who integrated an enhanced savings algorithm with bin packing heuristics. Building upon this foundation, Mohammad et al. [16] developed a hybrid method combining genetic algorithm (GA) for customer sequencing with simulated annealing for escaping local optima. Then Li et al. [17] introduced an ant colony algorithm with local preference adjustment, whose efficiency decreases with larger problem instances. Beyond these single-objective approaches, a small number of researchers have addressed multi-objective MTVRP through different methodologies, such as the hybrid heuristic algorithm proposed by Luo et al. [18] and the tabu search algorithm developed by Hong et al. [19]. Nevertheless, these MTVRP approaches fail to achieve balanced robot scheduling and demonstrate significant limitations in addressing multi-objective optimization problems.

B. MRTA Approaches

MRTA problems in real-world scenarios often involve multiple conflicting objectives, such as minimizing total travel distance, balancing workload distribution, and optimizing resource utilization. To address these challenges, researchers have developed various sophisticated multi-objective approaches. Xue et al. [20] proposed a hybrid competitive optimization algorithm featuring adaptive grid partitioning for large-scale many-objective MRTA. Wei et al. [21] advanced the field with a multi-objective particle swarm optimization approach incorporating Pareto front (PF) refinement and probability-based leader selection. The integration of lin–kernighan–helsgaun [22] for pre-generating promising solutions in multi-objective evolutionary algorithms (MOEAs) represents another significant advancement. Additionally, Wen et al. [23] have introduced an indicator-based MOEA with hybrid encoding. However, the multi-trip characteristic in multi-robot transportation systems, combined with the specific requirements of travel constraints in real-world logistics, necessitates new algorithmic solutions beyond existing MRTA approaches.

C. MRRS Approaches

Research on MRRS remains relatively sparse. Dai et al. [8] pioneered this area by proposing a multi-objective discrete artificial bee colony (MODABC) algorithm for route construction and allocation among multiple harvesting robots. They also adapted NSGA-II [24] and MOEA/D [25] as comparative algorithms. Based on these, Guo et al. [12] developed a collaborative discrete artificial bee colony (CDABC), which enhances local search capabilities by establishing neighborhoods based on key robots. Dong et al. [26] introduced an effective MOEA (AMOEA) for routing and scheduling among robots, featuring a crossover strategy informed by non-dominated solution set information for global search, and iterative greedy thinking for local search enhancement. Wang et al. [27] investigated collaborative routing protocols for heterogeneous robotic fleet management. Kang et al. [28] presented a multi-objective teaching-learning-based optimization algorithm to address the multi-robot planning and dispatching problem.

These algorithms optimize solutions through the collective evolution of multiple individuals within a population. Each solution has consistent dimensions, which imposes certain limitations on their representation. Moreover, these algorithms conduct searches based on the complete solution, thereby exhibiting limited capability for local exploration of different routes within a solution. To tackle these challenges, RNSGA is proposed by integrating variable neighborhood search

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theory [29] (which is good at performing local exploration) with population-based ideology [30] (which is skilled at diversity search in multi-objective optimization).

III. Problem Modeling

This research aims to address the MRRS problem via establishing the mathematical model.

A. Robot Setting

Without loss of generality, the robots are designed as follows:

- All robots are identical, fully automatic [31], functioning normally, and treated as integrated units [32].
- Once a robot reaches full capacity, it must return to the warehouse to unload.
- The speed v of robots is influenced by their drive power Dp, self-weight of the robot Sw, current load of the robot w. Then v can be roughly estimated by:

$$v = \frac{Dp}{\mu \times (Sw + w) \times (g)} \tag{1}$$

where friction coefficient μ is a constant, set to 0.1 in this study [33]. g represents the coefficient of gravity.

B. Problem Formulation

Consider a multi-robot logistic scenario where collection task points are distributed throughout the environment. These task points correspond to coordinates (y) where operations must be performed. The depot is randomly situated within the scene, and the process where a robot departs from the depot, completes multiple tasks, and returns to unload is defined as completing one route. Each route comprises a sequence of task points that determines the operational order. Since this study focuses solely on the transportation model, we exclude the specific task operations and their energy consumption but incorporate the execution time to better optimize routing schedules. As illustrated in Fig. 1, $\{8, 5, 10\}$ represents a possible route. The time required to complete a route is termed 'route duration', the energy expended is referred to as 'route energy consumption', and the combined weight of items collected along a route (Wt) is designated as the 'task amount' for that route. The calculation of *time* and *energy* are as follows:

$$\begin{aligned}
time_i &= t1_i + t2_i + t3_i \\
t1_i &= \sum_{j=0}^{k_i} \frac{y_{j+1} - y_j}{v_j} \\
t2_i &= Wt_i \\
t3_i &= Wt_i * 0.5 \\
Wt_i &= \sum_{j=1}^{k} \frac{Yield_j}{P}, \quad i = \{1, 2, \ldots, TN\} \\
energy_i &= t1_i \times Dp
\end{aligned} \tag{2}$$

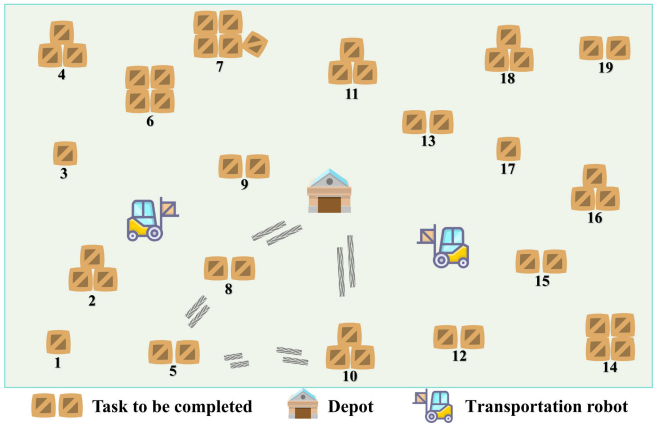


Fig. 1. Schematic diagram of the working scenario.

where *Yield* and P represent the task amount at each task point and the completion rate per unit task, respectively. To reduce the complexity of the results, this paper simplifies the model by setting the average weight of each pickup task as 1-kilogram with a 1-second collection time. These parameters can be adjusted according to practical requirements without affecting the algorithm's applicability designed in Section IV. Eq. (2) specifies that a route duration includes the time for robot path traversal *t1*, the operation time *t2*, and unloading time *t3*. When calculating *t1*, both y_0 and y_{k_i+1} correspond to the coordinates of the depot and v_0 corresponds to a robot's speed without load.

Each robot may need to perform multiple routes, forming a set of tasks (*TS*) for the robot. Since every route has the same starting and ending points (the depot), there is no sequential order among routes in the *TS*. The total time consumption (*TIME*) and total energy consumption (*ENERGY*) of all routes in a *TS* are obtained in Eq. (3):

$$\begin{aligned}
TIME_i &= \sum_{j=1}^{N_i} time_j, \quad (i = 1, 2, \ldots, r) \\
ENERGY_i &= \sum_{j=1}^{N_i} energy_j, \quad (i = 1, 2, \ldots, r)
\end{aligned} \tag{3}$$

where N is the number of routes in a *TS*. The relationship among task, route, and *TS* is detailed in the solution representation in Section IV-A.

The work aims to find a solution for MRRS that satisfies the following objectives:

1. Minimize total energy consumption as f_1:

$$\min TEC = \min \sum_{j=1}^{r} ENERGY_j \tag{4}$$

2. Minimize completion time of the most time-consuming robot as f_2:

$$\min MCT = \min \max(TIME_i), \quad (i = 1, 2, \ldots, r) \tag{5}$$

The f_1 aims to minimize the total energy consumption (*TEC*) of all robots. When addressing practical problems, f_1 should

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be customized based on the specific characteristics of the robots. The f_2 calculates the minimum completion time of the most time-consuming robot (MCT).

The following constraints need to be met when solving the problem:

1. The capacity of the robot should never be greater than its upper limit Wm:

$$Wt_i \leq Wm \quad (i = 1, 2, \ldots, TN) \tag{6}$$

2. Each task needs to be completed and only once:

$$t = \sum_{j=1}^{TN} k_j \tag{7}$$

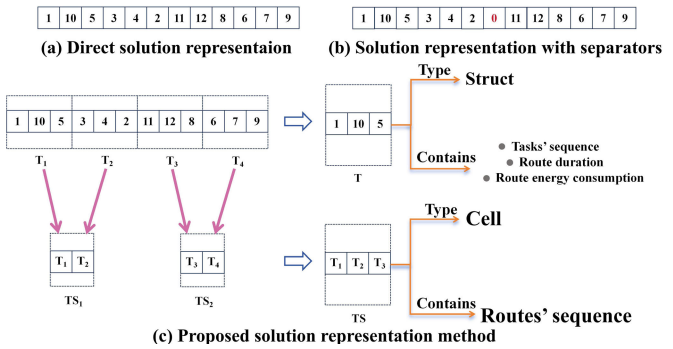


Fig. 2. Solution representation.

C. Problem Parameters

Due to the complexity of the problem, a substantial number of parameters are used in the problem formulation and algorithm design processes. To avoid confusion, we summarize the frequently used parameters throughout this paper as follows:

n	The total number of task points in the scene.
r	The number of robots.
y	Coordinates of task points.
v	Average moving speed on a travel between two task points.
Dp	The driving power of the robot, whose maximum is 2 kilowatt in this paper.
$Yield$	The task amount at each task point.
Wm	Maximum capacity of a robot.
Sw	Self-weight of a robot, which is set to half of Wm.
w	Robot's real-time load.
T	A route which consists of multiple task points.
TS	A task set, which consists of multiple routes.
AT	A collection of all routes.
Wt	The task amount of a route.
k	The number of task points in a route.
N	Number of routes in a TS.
TN	Total number of routes constructed in the scene.
$time$	Route duration.
$energy$	The route energy consumption.
$TIME$	Total time consumption of a robot.
$ENERGY$	Total energy consumption of a robot.

IV. Proposed Algorithm

To better address the MRRS problem, we propose RNSGA algorithm that consists of two key steps such as route construction and route allocation. The role of route construction is to determine the task points included in each route and establish the sequence in which these points will be visited, ensuring rational route structures. This step contains solution initialization and neighborhood search with route balance mechanism (RBM) and proximity-based optimization mechanism (PBOM). All of these processes need to optimize the sequence of task points with intro-route sequence adjustment (ISA). Subsequently, route allocation is performed, where population-based route allocation mechanism (PRAM) is employed to determine which routes each robot needs to perform.

The following subsection provides a brief overview of how solutions are represented. After that, we outline the overall framework of the algorithm. Finally, the last two subsections detail the route construction and route allocation, respectively.

A. Solution Representation

Fig. 2 illustrates three methods for representing solutions, where (a) and (b) are commonly used in existing population-based optimization algorithms, and (c) is the representation proposed in this study. Fig. 2(a) uses a direct solution expression scheme to represent solutions [34], maintaining sequence simplicity. The sequence represents the order of execution for tasks. However, it cannot directly illustrate the route allocation mode. Therefore, it requires many archives to represent routes and TSs. Fig. 2(b) addresses this issue via dividing the TSs by inserting '0' elements in the sequence [8]. Nevertheless, the number of routes in each TS and how many task points in each route must be evaluated during function assessment. Moreover, since the content of TSs corresponding to different solutions may vary, updating solutions during population updates poses challenges for crossover and mutation. In summary, these two methods of solution representation increase workload and complexity greatly.

To clearly divide routes, we use programming logic combined with problem-specific characteristics to represent solutions using tuples and structs, as depicted in Fig. 2(c). The encoding structure [35] is designed as a three-level hierarchy: at the lowest level, task points are represented by numerical sequences indicating both their index and completion order, like Fig. 2(a). At the middle level, each route T is encoded as a structured unit containing task sequences, duration $time$ and energy consumption $energy$ of this route. Each route represents a complete cycle where a robot departs from the depot, executes the designated tasks in sequence, and returns. At the highest level, routes are numbered and integrated into TSs, allowing for intuitive representation of each robot's assignments.

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Algorithm 1 General Framework of RNSGA

Input: Coordinates of task points y, task amount at each task point *Yield*, maximum capacity of a robot *Wm*, population size *Pnum*

***Output*:** Non-dominated solutions $Archive$ and their corresponding fitness value *Fit*

```
Parameter initialization
tic // Start timing
while termination criterion not fulfilled do
    if in the first loop then
        Generate an initial solution AT // Algorithm 2
        AT ← ISA(AT) // Optimize AT with Algorithm 3
        ARCHIVE ← ∅
        Non ← AT // Initialize the non-dominated solution
    end if
    archive_1 ← RBM(Non) // Algorithm 4
    archive_1 ← ISA(archive_1) // Only optimize the adjusted T
    [archive_1, FIT] ← PRAM(archive_1) //Algorithm 7
    [archive_1, Fit] ← EnvironmentalSelection(archive_1, FIT, Pnum)
    archive_2 ← PBOM(archive_1) // Algorithm 6
    archive_2 ← ISA(archive_2) // Only optimize the adjusted T
    if archive_1 = archive_2 then
        archive_2 ← Perturbation(archive_2) // Randomly select some routes to reinitialize
    end if
    [archive, FIT] ← PRAM(archive_1 ∪ archive_2) //Algorithm 7
    [ARCHIVE, Fit] ← EnvironmentalSelection(ARCHIVE ∪ archive, FIT, Pnum)
    Non ← ARCHIVE // Obtain the non-dominated solutions
    toc // Judging termination criterion
end while
```

By using this representation of the solution, the objective function can be reconstructed to reduce the workload of each function evaluation. The total energy consumption and completion time of each robot can be calculated as the sum of the time and energy values of all its executed routes. Only the updated routes need to be evaluated while reusing the time and energy consumption values of unchanged routes. This enables separate optimization of T or TS, thereby improving computational efficiency.

B. Algorithm Structure

The implementation of RNSGA is detailed in Algorithm 1. And the main procedures are briefly summarized as follows.

1) Parameter initialization: load the primary parameters provided in the problem.

2) Once the terminal criterion is not fulfilled, the algorithm enters the main loop.

3) Initial solution generation: initialize the routes in the first loop. Through this step, a complete initial solution is obtained.

4) Intro-route sequence adjustment: optimize each route in the solution with ISA.

5) Route balance: execute the RBM mechanism on the non-dominated solutions in each iteration. And the ISA is reapplied to the modified routes.

6) Route allocation: reassign all routes to the available robots in each solution to generate new solutions and obtain their corresponding objective values.

7) Proximity-based optimization: apply PBOM to optimize solutions sequentially. Similarly, the ISA is employed to further optimize the modified routes, followed by route reallocation.

The general framework is also depicted in Fig. 3. During these operations, multiple solutions may arise. However, for ease of representation, Fig. 3 only illustrates the transformation process of one solution. Due to page limitations, Algorithms 2-7 are shown in the supplementary materials in Section S-II.

C. Route Construction

Route construction is employed to establish the sequence of task points within each route. In this step, a clustering-based initialization method is proposed to generate high-quality initial solutions. ISA is designed to optimize task sequence within each route. RBM and PBOM are proposed to further optimize task point assignments, balancing resource allocation and reducing redundant paths respectively.

1) Solution initialization method: High-quality initial solutions contribute to rapid algorithm convergence and good results. This paper proposes a novel solution initialization method for route initialization based on problem characteristics and solution representations.

Fig. 4 illustrates the process of initialization. Colored circles represent pending tasks, which are randomly distributed throughout the scene. Initially, all tasks are clustered, with the number of clusters equal to r. Each cluster represents a *TS*. For example, with three robots, all pending tasks are grouped into three clusters, each represented by a different color in Fig. 4(a). The process then repeats as follows: Select the cluster closest to the depot. Each route within the set is

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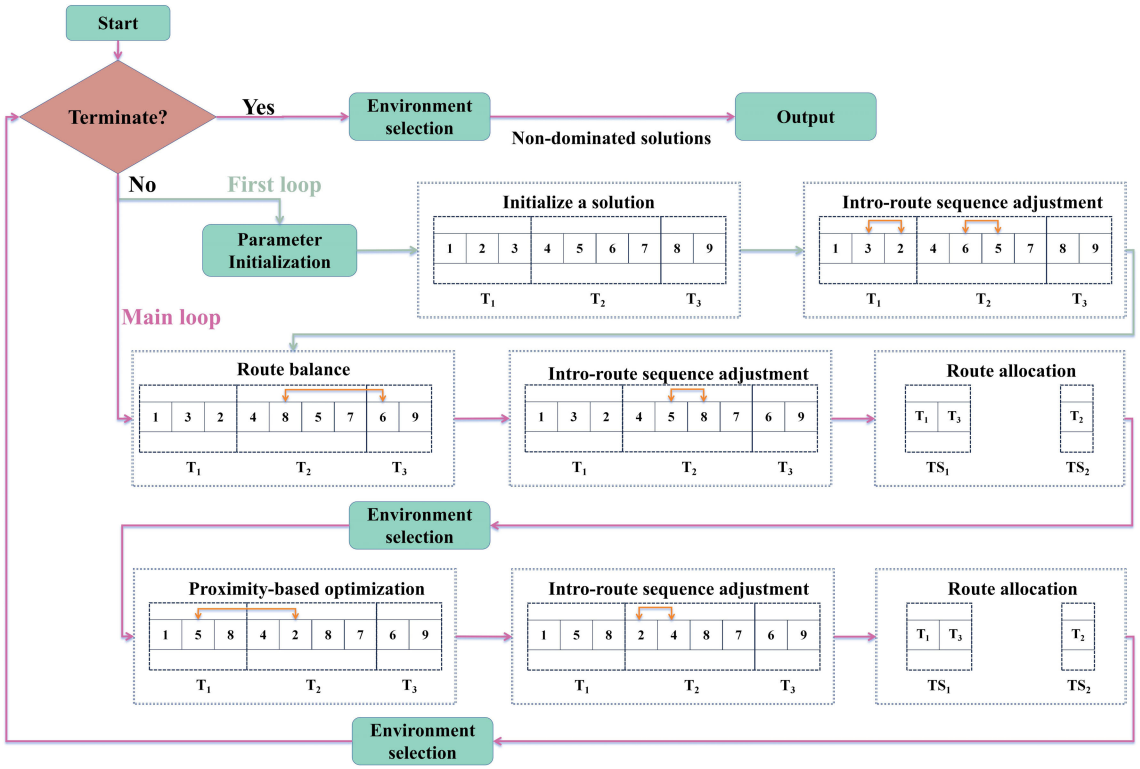


Fig. 3. General framework of RNSGA.

initialized individually. First, an empty route is constructed. Then the task point closest to the depot is selected to fill it in. Subsequently, the sequence is filled sequentially with the nearest task point to the last one in the sequence (nearest-task principle), ensuring that points are selected exclusively from the current cluster. A sequence is initially formed when adding any complete task would cause robot overload, which is referred to as critical state (*CS*). This process continues until the remaining points in the current cluster are insufficient to reach the *CS*, as shown in Fig. 4(b). At this point, the remaining nodes are mixed with all other remaining tasks to form $r-1$ clusters anew, depicted in Fig. 4(c). This process repeats until all *TS*s are constructed, as depicted in Fig. 4(d). The detailed process is shown in Algorithm 2.

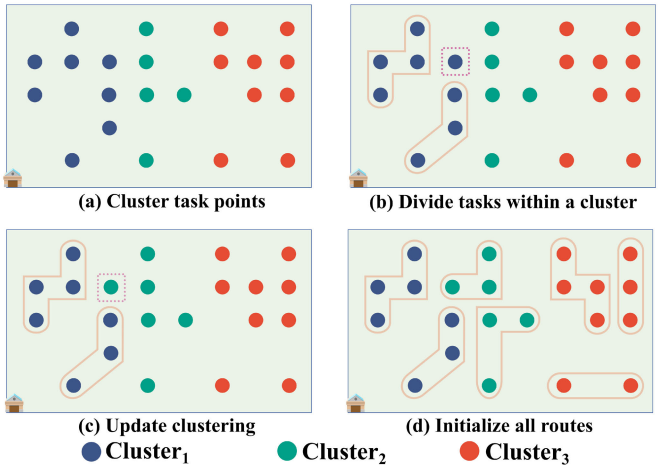


Fig. 4. Solution initialization.

2) Intro-route sequence adjustment: In the aforementioned initialization mode, the task sequences within each route are ordered based on the nearest-task principle. Thus, this initialization method serves only to preliminarily define the tasks within each route without further exploring their ordering. Therefore, to further enhance the quality of the initialization results, it is necessary to optimize the initial solutions. We propose an improved discrete GA (IDGA) to perform ISA:

Assuming Fig. 5(a) represents the initialization result. Taking the labeled red task sequence as an example: according to the nearest-task principle, the initial route is $1, 2, 3, 4$. Additionally, a randomly generated sequence is merged with the original sequence to form a new population. Subsequently, the following steps are repeated: Selecting the sequence with the shortest duration in the population as $gbest$. Then, using the newly generated population and updated $gbest$ with IDGA to produce offspring, as depicted in Fig. 5(b).

On the basis of GA, IGDA contains a variety of discrete sequence crossover methods: partially-matched crossover, order-based crossover, and subtour exchange crossover [36]. In the early stage of optimization, all crossover methods are randomly adopted. As the search goes on, the probability $Prob$ of each crossover method being adopted are adjusted adaptively according to the number of successful times Suc of sequence optimization:

$$Prob_i = \frac{Suc_i}{\sum_{j=1}^{3}(Suc_j)} \quad \forall i \in \{1, 2, 3\} \tag{8}$$

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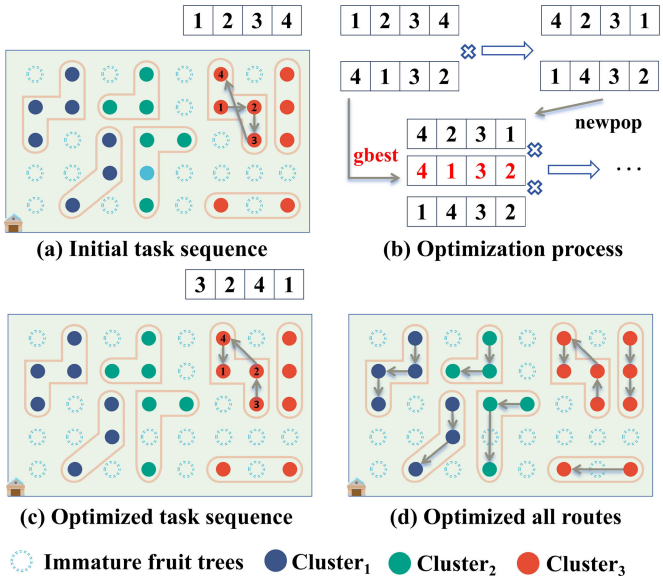


Fig. 5. Intro-route sequence adjustment.

With IDGA, the final result is the task sequence with the shortest *time*, as shown in Fig. 5(c). Similarly, the *time* for all task sequences are optimized in Fig. 5(d). The process is shown in Algorithm 3. Besides the solution initialization, any change made to any task within a route during subsequent operations of the RNSGA requires ISA to maximize the functionality of each operation as much as possible.

3) Route balance mechanism: The task amount for most routes obtained by initialization is similar. However, since the initial route count may not be a perfect multiple of the number of robots, some robots may end up idle. To further optimize the route structure, this paper proposes the RBM, as shown in Algorithm 4.

Firstly, based on *Yield*, *Wm*, r, and *TN*, the number z of idle routes is calculated according to the current initialization principles. For instance, in Fig. 6(a), assuming 14 tasks are initialized across 3 robots into 5 routes. To ensure each robot performs an equal number of routes, z is set to 1. Idle routes are represented as 0 in Fig. 6(b).

Next, the most time-consuming route (*MTC*) and the least time-consuming route (*LTC*) are identified among all routes. When their time difference exceeds a threshold (THD), their routes need balancing, where THD is equivalent to the minimum task quantity among all tasks encompassed within the *MTC*.

The ISA operation results in a closely interconnected sequence of tasks. Hence, tasks at the head or tail of the sequence are relatively distant from the route center. Therefore, priority is given to transferring *MTC*'s head or trail to *LTC*:

- Starting from the *MTC* and *LTC*, transfer *MTC*'s leading tasks to *LTC* one by one until their combined route duration no longer decreases, as shown in Fig. 6(c).
- Starting from the original *MTC* and *LTC*, transfer *MTC*'s trailing tasks to *LTC* one by one until their combined route duration no longer decreases, as shown in Fig. 6(d).

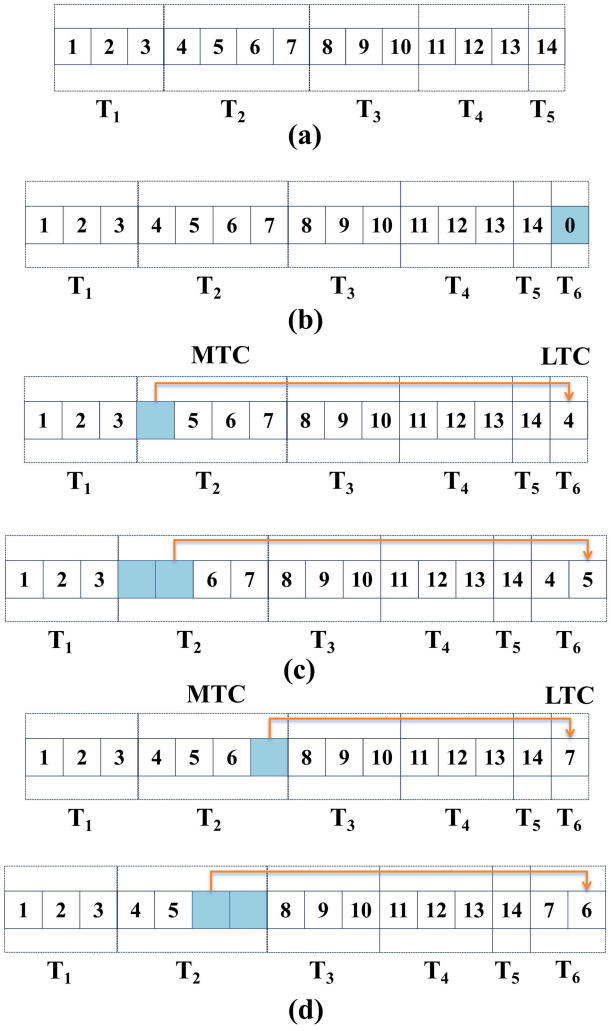


Fig. 6. Route balance mechanism.

These two processes are referred to as the exchange operation, described in Algorithm 5. It generates a series of corresponding *MTC* and *LTC* pairs, which are then integrated into the original solution to produce a series of new solutions. Since other routes in the original solution remain unchanged, the total duration change of *MTC* and *LTC* corresponds directly to the *TEC*, highlighting the advantage of this solution representation method. The solution with the smallest *MCT* is chosen, *MTC* and *LTC* are updated, and the exchange operation is repeated until the difference between *MTC* and *LTC* is below THD.

After removing duplicate results, ISA is needed to optimize the adjusted routes in each solution. These solutions are then updated through the execution of PRAM, which is described in Section IV-D. From a comprehensive perspective, RBM is designed to balance route durations across different robots, thereby placing greater emphasis on *MCT* optimization.

4) Proximity-based optimization mechanism: As mentioned above, ISA can only optimize the sequence order. Additionally, RBM only reallocates tasks within *MTC* and *LTC*, without further optimization for other routes. Thus, significant

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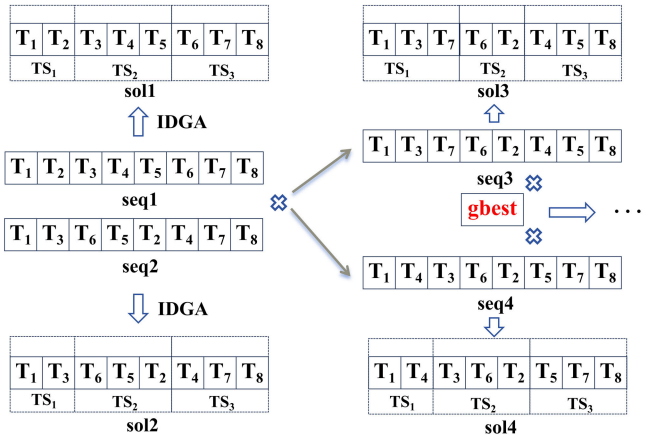


Fig. 7. Population-based route allocation mechanism.

optimization opportunities remain in these routes. To address these issues, this paper introduces PBOM, as shown in Algorithm 6.

PBOM takes the optimal solution that RBM obtained as input. Initially, the central coordinates of each route in this solution are computed to facilitate route distribution assessment. Subsequently, a neighborhood is randomly selected to choose a route (op_1) to be optimized. The neighborhoods contain:

- N_1: Select *MTC* as op_1 to optimize *MTC* and its neighbors.
- N_2: Select *LTC* as op_1 to optimize *LTC* and its neighbors.
- N_3: Randomly select a route as op_1 to perform diversity search.

Distances from op_1's central coordinate to those of other routes are calculated and sorted in ascending order, aiming to get a ranking. Then, routes are sorted by decreasing route energy consumption to obtain another ranking. Subsequently, the two rankings are combined to yield a comprehensive ranking. The top-ranked route is deemed the most cost-effective, facilitating a knowledge-guided local search (KGLS) [37] with op_1 to generate a series of new solutions. This process repeats as op_1's route duration or *TEC* continue to decrease. A simple example is shown in Section S-V in the supplementary materials.

PBOM serves to mitigate potential flaws from initialization and RBM. It eliminates redundant paths, significantly reducing the proportion of robot movement time in optimized routes. Furthermore, by optimizing routes' structure, it can decrease the workload of neighboring high-duration routes, thereby enhancing work efficiency. From a comprehensive perspective, while PBOM is capable of optimizing both objective values concurrently, it demonstrates particularly notable effectiveness in improving TEC.

D. Route Allocation

Route allocation serves to redistribute the optimized routes among robots, thereby achieving solution updates. Random allocation introduces significant uncertainty and makes it challenging to ensure the quality of the assignments. Therefore, we employ PRAM to perform this operation.

PRAM optimizes at the route level, without adjusting the tasks further. Thus, the *TEC* value of the routes remain unchanged throughout the process. PRAM's role is to allocate a route sequence to r robots to achieve a multi-robot route allocation solution with the minimum *MCT* value, as described in Algorithm 7.

Let the route indices in *AT* correspond to $T_1, T_2, \ldots, T_{TN}$. First, two route index sequences, $seq1$ and $seq2$, are randomly generated. Then, using IDGA, breakpoints are applied to these index sequences to allocate *TN* routes to r robots. To facilitate the presentation, Fig. 7 illustrates the process of assigning 8 routes to 3 robots. This yields allocation results *sol1* and *sol2*, along with work duration vectors *TIME1* and *TIME2* for allocation results separately. Hence, the *MCT* values of *sol1* and *sol2* are determined by calculating the maximum values of *TIME1* and *TIME2* respectively. All newly generated index sequences are then stored in a population, and the sequence with the smallest corresponding *MCT* value is selected as $gbest$. Next, using IDGA in conjunction with $gbest$ and the newly generated population, new sequences are produced. This process is repeated until the population size reaches *Pnum*. Finally, the solution with the smallest *MCT* value is chosen as the optimized result for PRAM on the current *AT*.

V. Experimental Results and Analysis

To rigorously assess the effectiveness of the proposed RNSGA algorithm, we employ a comprehensive evaluation framework incorporating design of experiments (DOE) methodology and the Wilcoxon rank-sum statistical test. These analytical approaches have been extensively validated in evaluating combinatorial optimization algorithms [38] and conducting algorithmic performance comparisons [39], [40], [41]. This section systematically presents the experimental setup, comparative performance analysis using quality indicators, analysis of default solution and case visualization, and other experiments that demonstrate the algorithm's capabilities.

A. Experimental Setup

To validate the competitiveness of our proposed algorithm in transportation systems, we employ the widely-recognized benchmark [16] data-set P [42] as our testing scenario. As detailed in Table I, the problem instances are characterized by several key parameters: n and k in the instance name denote the number of nodes and required robots, respectively. The spatial distribution of demand points is characterized by μ_y and λ_y, representing the average and maximum distances from task points to the depot, respectively. The demand intensity at each node is quantified by μ_d (average demand) and λ_d (maximum demand). The parameter Q specifies the maximum capacity constraint for each robot in the fleet.

Given the inherent conflicting nature between the two objectives in this study, direct comparison of objective values proves inadequate for comprehensive performance evaluation [43]. Therefore, we employ two widely-adopted

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TABLE I
SPECIFICATIONS FOR THE EXPERIMENTAL PROBLEMS

Problems	μ_y	λ_y	μ_d	λ_d	Q
P-n16-k8	25.328	32.5576	15.375	31	35
P-n19-k2	26.6635	43.9318	16.3158	31	160
P-n20-k2	26.8869	43.9318	15.5	31	160
P-n21-k2	26.0426	43.9318	14.1905	30	160
P-n22-k2	26.0488	43.9318	14	30	160
P-n22-k8	27.7502	49.366	1022.7273	2500	3000
P-n23-k8	24.9664	43.9318	13.6087	30	40
P-n40-k5	23.8694	43.9318	15.45	41	140
P-n45-k5	24.8765	43.9318	15.3778	41	150
P-n50-k10	22.7812	37.108	19.02	37	100
P-n50-k7	22.7812	37.108	19.02	37	150
P-n50-k8	22.7812	37.108	19.02	37	120
P-n51-k10	24.0235	43.9318	15.2353	41	80
P-n55-k10	22.6081	37.108	18.9455	37	115
P-n55-k15	22.6081	37.108	18.9455	37	70
P-n55-k7	22.6081	37.108	18.9455	37	170
P-n55-k8	22.6081	37.108	18.9455	37	160
P-n60-k10	23.4909	42.4264	18.9	37	120
P-n60-k15	23.5088	42.4264	18.9	37	80
P-n65-k10	24.254	43.2666	18.7538	37	130
P-n70-k10	24.205	43.2666	18.7571	37	135
P-n76-k4	24.2057	43.2666	17.9474	37	350
P-n76-k5	24.2057	43.2666	17.9474	37	280
P-n101-k4	24.9471	49.93	14.4356	41	400

performance metrics for multi-objective optimization, the Hypervolume (HV) [44] and the Inverted Generational Distance (IGD) [45], to assess the algorithms' performance. The computation of these metrics necessitates the establishment of distinct reference points for each problem instance, ensuring a fair and systematic comparison.

Point (1, 1) is selected as the reference point in each test instance [8] for calculating HV value:

$$\mathrm{HV}(P) = \delta(\cup_{i=x}^{|P|} v_x) \tag{9}$$

where P is a obtained non-dominated solution set of a test instance and $|P|$ represents its number. δ denotes the Lebesgue measure for volume calculation, while v_x indicates the hypervolume bounded by the xth solution and the reference point within the solution set. Therefore, HV quantifies the volume of the hypercube defined by the solution set and reference points within the objective space. The farther a result is from the reference point within a problem, the bigger HV value and the better the quality of the result.

Subsequently, for each comparative experiment, we first need to tally the output results of all compared algorithms involved in each test instance of each problem. Removing duplicate values gives us the optimized results, referred to as $optimal$. Next, we compute the non-dominated solutions in each set of $optimal$, resulting in the reference points P^* for calculating IGD values in each problem:

$$\mathrm{IGD}(P, P^*) = \frac{\sum_{x \in P^*} d(x, P)}{|P^*|} \tag{10}$$

where $d(x, P)$ computes the Euclidean distance from point $x \in P^*$ to its nearest neighbor in set P. Consequently, IGD provides a metric for the average proximity between the true Pareto optimal front and the non-dominated solutions obtained by the algorithm. The smaller the IGD value, the better the result.

The computational experiments are subject to a dynamic termination criterion based on maximum CPU runtime, defined as Ft_{max}=n. The final Pareto-optimal solutions are maintained at a consistent population size of *Pnum*. All algorithmic implementations are executed on a workstation configured with an Intel Core i7-12700 processor (2.1GHz) and 32GB of RAM.

B. Comparative Performance Analysis Using Quality Indicators

To verify the effectiveness of RNSGA, it is compared with six representative algorithms: AMOEA [26], CDABC [12], MODABC [8], MOEA/D [25], NSGA-II [24], and MOTS [46]. Among them, AMOEA, CDABC, and MODABC are algorithms proposed in these two years for solving the MRRS problem. MOEA/D and NSGA-II are highly representative MOEAs, and in this study, their solution representation, initialization modes, and crossover operation are consistent with those of MODABC. MOTS, a classic multi-objective tabu search method, has significantly influenced the design of RNSGA with its approach of using single points separately to search multi-objective optimization problems. Its solution representation and initialization modes are consistent with RNSGA, with the size of the candidate solution archive set to *Pnum*. The experimental settings for the comparative experiments are consistent with those in Section V-A, with other algorithms using their default settings.

The final test results of 24 scenarios are shown in Table II. In the tables, '+', '-', and '=' indicate that the comparison algorithm's results are significantly better than, worse than, or equal to the results obtained by RNSGA, respectively. From the experimental results, RNSGA is clearly superior to the other algorithms. Furthermore, several representative test instances are selected to present all the non-dominated solutions obtained by algorithms in Fig. 8. The solutions obtained by RNSGA (denoted by purple asterisks) consistently occupy the first layer of the PF, which further validates the superior performance of RNSGA. Below, we select some major algorithms for detailed experimental analysis.

NSGA-II and MOEA/D demonstrate stable performance, primarily attributable to their streamlined yet efficient framework designs. NSGA-II maintains solution diversity through its non-dominated sorting, demonstrating particular efficiency in allocating independent routes for customer nodes with high demand volumes. Meanwhile, MOEA/D achieves excellent balance through its decomposition-based approach. Both algorithms successfully avoid the pitfalls of excessive dependence on randomized selection mechanisms.

Conversely, MODABC's excessive random search mechanisms compromise its performance. While MODABC's three-phase structure (employed bee, onlooker bee, and scout bee phases) collaboratively conducts diversified exploration in the objective space, its effectiveness diminishes as problem dimensions increase. When both task points and robot number expand, the limitations of its random selection mechanism become evident: the algorithm's random strategy for selecting which *TS* to optimize often results in critical paths

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TABLE II
COMPREHENSIVE RESULTS WITH QUALITY INDICATORS

IGD

Problems	AMOEA [26]	CDABC [12]	MODABC [8]	MOEA/D [25]	MOTS [46]	NSGA-II [24]	RNSGA
P-n16-k8	1.9901e+01 (3.60e+00) -	5.8845e+00 (3.20e+00) -	6.4149e+00 (3.43e+00) -	6.5747e+00 (3.66e+00) -	6.1412e+00 (3.77e+00) -	6.0427e+00 (4.00e+00) -	**9.7347e-02 (5.07e-02)**
P-n19-k2	8.7791e+00 (1.03e+00) -	1.0034e+01 (1.36e+00) -	1.1422e+01 (8.24e-01) -	1.1997e+01 (9.41e-02) -	8.0353e+00 (2.32e+00) -	4.9704e+00 (6.51e-01) -	**3.8667e-01 (2.84e-01)**
P-n20-k2	4.7500e+01 (5.42e+00) -	2.8803e+01 (3.97e+00) -	2.8732e+01 (4.09e+00) -	2.4212e+01 (5.24e+00) -	9.8353e+00 (2.77e+00) -	7.2463e+00 (2.71e+00) -	**3.7738e-01 (2.29e-01)**
P-n21-k2	1.2741e+01 (1.87e-15) -	4.5398e+00 (7.08e-01) -	4.9616e+00 (4.30e-01) -	5.1178e+00 (9.36e-16) -	4.6101e+00 (1.30e+00) -	2.3496e+00 (1.37e+00) -	**1.3981e-01 (2.36e-01)**
P-n22-k2	1.4050e+01 (2.95e-01) -	8.0928e+00 (2.41e-01) -	8.0162e+00 (3.20e-01) -	7.8300e+00 (7.79e-01) -	6.9666e+00 (1.37e+00) -	4.3457e+00 (8.89e-01) -	**8.9082e-01 (4.59e-01)**
P-n22-k8	1.6491e+03 (3.32e+02) -	9.7581e+02 (3.83e+02) -	1.2918e+03 (3.96e+02) -	1.0555e+03 (1.65e+02) -	9.5150e+02 (1.80e+02) -	7.7384e+02 (2.53e+02) -	**9.7757e+01 (3.17e+01)**
P-n23-k8	1.1705e+01 (1.87e-01) -	1.0028e+01 (8.31e-01) -	1.0016e+01 (1.07e+00) -	9.4995e+00 (1.45e+00) -	9.7640e+00 (1.17e+00) -	7.7168e+00 (2.24e+00) -	**2.6566e+00 (1.12e+00)**
P-n40-k5	5.3894e+01 (7.49e-15) -	8.3371e+01 (4.28e+00) -	8.4671e+01 (2.21e+00) -	5.4050e+01 (2.12e+01) -	4.6918e+01 (4.41e+00) -	2.2244e+01 (4.09e+00) -	**4.1399e+00 (1.81e+00)**
P-n45-k5	7.1973e+01 (1.50e-14) -	9.0998e+01 (2.08e+00) -	9.1307e+01 (3.49e+00) -	7.4375e+01 (2.23e+01) -	5.1182e+01 (4.52e+00) -	2.3625e+01 (2.88e+00) -	**8.2749e+00 (4.68e+00)**
P-n50-k10	3.6600e+01 (7.49e-15) -	3.6159e+01 (1.20e-01) -	3.6209e+01 (6.47e-02) -	2.7979e+01 (4.30e+00) -	3.4508e+01 (4.95e+00) -	1.5108e+01 (2.13e+00) -	**5.7768e+00 (3.46e+00)**
P-n50-k7	3.2077e+01 (0.00e+00) -	6.1768e+01 (1.27e-01) -	6.1731e+01 (1.57e-01) -	5.0376e+01 (1.34e+01) -	5.0702e+01 (1.16e+01) -	2.4858e+01 (5.66e+00) -	**6.0525e+00 (1.14e+00)**
P-n50-k8	5.9694e+01 (1.75e-01) -	5.9210e+01 (5.97e-02) -	5.9247e+01 (3.96e-02) -	4.3397e+01 (1.12e+01) -	5.4732e+01 (1.86e+00) -	2.1249e+01 (4.40e+00) -	**6.0617e+00 (2.67e+00)**
P-n51-k10	6.6939e+01 (1.71e+01) -	6.0151e+01 (2.73e+00) -	6.1543e+01 (1.29e+00) -	3.2564e+01 (1.04e+01) -	3.9403e+01 (3.75e+00) -	1.8530e+01 (5.44e+00) -	**9.7236e+00 (1.55e+00)**
P-n55-k10	5.6962e+01 (7.26e+00) -	6.6927e+01 (6.69e-02) -	6.6980e+01 (8.46e-02) -	4.9023e+01 (1.62e+01) -	6.1727e+01 (9.59e+00) -	1.9709e+01 (3.72e+00) -	**4.8982e+00 (3.07e+00)**
P-n55-k15	2.8885e+01 (7.49e+00) -	5.2458e+01 (8.08e-02) -	5.2478e+01 (8.18e-02) -	2.9148e+01 (8.12e+00) -	4.8441e+01 (3.96e+00) -	2.0069e+01 (5.48e+00) -	**2.6419e+00 (1.02e+00)**
P-n55-k7	3.6335e+01 (7.49e-15) -	5.6718e+01 (2.12e-01) -	5.6892e+01 (2.37e-01) -	4.9229e+01 (6.84e+00) -	4.7066e+01 (1.06e+01) -	2.7247e+01 (3.74e+00) -	**1.0512e+01 (2.67e+00)**
P-n55-k8	1.0409e+02 (1.23e+00) -	6.3667e+01 (8.63e-01) -	6.3722e+01 (1.51e+00) -	5.8749e+01 (4.43e+00) -	5.6856e+01 (4.91e+00) -	2.7238e+01 (4.30e+00) -	**4.8071e+00 (2.12e+00)**
P-n60-k10	4.8710e+01 (7.49e-15) -	5.8291e+01 (3.89e-01) -	5.8585e+01 (5.34e-02) -	4.2441e+01 (9.95e+00) -	5.0783e+01 (1.17e+01) -	2.2477e+01 (3.31e+00) -	**4.4665e+00 (1.58e+00)**
P-n60-k15	5.3417e+01 (1.37e+00) -	7.0571e+01 (3.25e+00) -	7.2148e+01 (1.38e+00) -	3.5898e+01 (8.11e+00) -	5.7600e+01 (5.19e+00) -	2.4850e+01 (6.54e+00) -	**6.8479e+00 (1.57e+00)**
P-n65-k10	5.3586e+01 (7.49e-15) -	3.4764e+01 (1.30e+00) -	3.4982e+01 (1.35e+00) -	3.5578e+01 (1.91e-01) -	3.3077e+01 (1.42e+00) -	2.4229e+01 (3.05e+00) -	**9.0277e+00 (4.09e+00)**
P-n70-k10	8.3176e+01 (1.65e+00) -	5.8531e+01 (2.04e+00) -	5.8848e+01 (2.33e+00) -	4.8987e+01 (5.38e+00) -	4.6924e+01 (5.95e+00) -	2.7641e+01 (4.92e+00) -	**1.9536e+01 (2.58e+00)**
P-n76-k4	2.3603e+02 (0.00e+00) -	8.4377e+01 (5.15e+00) -	8.5875e+01 (4.51e+00) -	8.8810e+01 (2.97e+00) -	6.6259e+01 (7.17e+00) -	5.8514e+01 (6.66e+00) -	**1.6384e+01 (5.87e+00)**
P-n76-k5	1.0723e+02 (0.00e+00) -	5.8378e+01 (1.18e+00) -	5.9491e+01 (7.13e-01) -	5.8589e+01 (1.97e+00) -	5.5021e+01 (4.37e+00) -	4.2871e+01 (9.50e+00) -	**1.3060e+01 (3.65e+00)**
P-n101-k4	1.6308e+02 (3.00e-14) -	1.1873e+02 (3.41e+00) -	1.2098e+02 (1.27e+00) -	1.0536e+02 (8.83e+00) -	9.9960e+01 (9.87e+00) -	6.8363e+01 (8.24e+00) -	**1.2588e+01 (8.25e+00)**
+/-/=	0/24/0	0/24/0	0/24/0	0/24/0	0/24/0	0/24/0	

HV

Problems	AMOEA	CDABC	MODABC	MOEA/D	MOTS	NSGA-II	RNSGA
P-n16-k8	1.1534e-01 (1.28e-02) -	1.5298e-01 (1.01e-02) -	1.5020e-01 (9.90e-03) -	1.5968e-01 (1.14e-02) -	1.5391e-01 (1.02e-02) -	1.7025e-01 (1.59e-02) -	**2.2728e-01 (1.10e-04)**
P-n19-k2	2.1528e-01 (4.45e-03) -	2.1778e-01 (6.49e-03) -	2.1048e-01 (3.85e-03) -	2.1098e-01 (2.34e-03) -	2.2807e-01 (7.55e-03) -	2.5234e-01 (1.94e-03) -	**2.6089e-01 (1.73e-03)**
P-n20-k2	1.3055e-01 (6.50e-03) -	1.8553e-01 (4.13e-03) -	1.8548e-01 (3.86e-03) -	1.8545e-01 (1.67e-02) -	2.1312e-01 (4.62e-03) -	2.2243e-01 (3.17e-03) -	**2.3809e-01 (6.66e-04)**
P-n21-k2	1.9549e-01 (0.00e+00) -	2.2810e-01 (2.94e-03) -	2.2635e-01 (1.78e-03) -	2.2571e-01 (2.93e-17) -	2.2817e-01 (5.60e-03) -	2.4112e-01 (2.76e-03) -	**2.4877e-01 (3.70e-04)**
P-n22-k2	1.4997e-01 (1.00e-03) -	1.6979e-01 (1.05e-03) -	1.7012e-01 (1.39e-03) -	1.7544e-01 (2.10e-03) -	1.7763e-01 (3.05e-03) -	1.8906e-01 (3.32e-03) -	**1.9972e-01 (7.90e-04)**
P-n22-k8	1.5360e-01 (9.48e-03) -	1.7677e-01 (1.60e-02) -	1.6209e-01 (1.07e-02) -	1.8775e-01 (1.32e-02) -	1.9072e-01 (1.20e-02) -	2.2099e-01 (1.63e-02) -	**2.7246e-01 (4.20e-03)**
P-n23-k8	1.1172e-01 (1.31e-03) -	1.4278e-01 (4.20e-03) -	1.4256e-01 (7.69e-03) -	1.4406e-01 (6.52e-03) -	1.4511e-01 (5.83e-03) -	1.6499e-01 (1.11e-02) -	**2.1974e-01 (1.23e-02)**
P-n40-k5	1.6947e-01 (8.49e-03) -	1.6281e-01 (1.08e-02) -	1.6021e-01 (4.40e-03) -	2.0082e-01 (2.58e-02) -	2.1480e-01 (6.71e-03) -	2.6018e-01 (1.07e-02) -	**3.0354e-01 (4.61e-03)**
P-n45-k5	1.1492e-01 (1.47e-02) -	1.2452e-01 (1.29e-02) -	1.2024e-01 (1.33e-02) -	1.5472e-01 (3.63e-02) -	1.9854e-01 (9.74e-03) -	2.3999e-01 (5.76e-03) -	**2.8126e-01 (2.49e-03)**
P-n50-k10	2.4868e-01 (4.47e-03) -	2.6861e-01 (6.44e-03) -	2.6860e-01 (7.22e-03) -	2.8737e-01 (7.41e-03) -	2.7754e-01 (1.17e-02) -	3.2491e-01 (6.95e-03) -	**3.4082e-01 (1.27e-02)**
P-n50-k7	1.9937e-01 (2.93e-17) -	1.7002e-01 (5.20e-03) -	1.6865e-01 (5.00e-03) -	1.9628e-01 (1.89e-02) -	2.0592e-01 (6.78e-03) -	2.3516e-01 (7.76e-03) -	**2.7555e-01 (5.71e-03)**
P-n50-k8	1.6778e-01 (1.04e-02) -	1.7372e-01 (5.96e-03) -	1.7133e-01 (9.60e-03) -	2.1741e-01 (1.23e-02) -	2.1255e-01 (8.48e-03) -	2.5563e-01 (1.19e-02) -	**2.9159e-01 (7.27e-03)**
P-n51-k10	1.1806e-01 (3.02e-02) -	1.7987e-01 (1.23e-02) -	1.7272e-01 (6.60e-03) -	2.3266e-01 (2.02e-02) -	2.2718e-01 (1.02e-02) -	2.6791e-01 (1.94e-02) -	**3.1348e-01 (5.60e-03)**
P-n55-k10	1.4529e-01 (2.09e-02) -	1.6658e-01 (9.05e-03) -	1.6156e-01 (4.74e-03) -	1.9403e-01 (2.39e-02) -	1.8948e-01 (9.59e-03) -	2.4607e-01 (5.43e-03) -	**2.7295e-01 (8.21e-03)**
P-n55-k15	1.5812e-01 (1.24e-02) -	1.2229e-01 (6.43e-03) -	1.2020e-01 (6.68e-03) -	1.7311e-01 (1.47e-02) -	1.4273e-01 (1.20e-02) -	2.0712e-01 (1.39e-02) -	**2.5458e-01 (5.27e-03)**
P-n55-k7	2.0986e-01 (0.00e+00) -	1.9437e-01 (3.63e-03) -	1.9102e-01 (2.18e-03) -	2.1040e-01 (1.28e-02) -	2.1651e-01 (6.72e-03) -	2.4818e-01 (8.50e-03) -	**2.8723e-01 (5.72e-03)**
P-n55-k8	1.1470e-01 (1.54e-02) -	1.8214e-01 (6.34e-03) -	1.7979e-01 (5.49e-03) -	2.0015e-01 (7.42e-03) -	2.0437e-01 (8.78e-03) -	2.5280e-01 (9.81e-03) -	**3.0565e-01 (4.37e-03)**
P-n60-k10	1.2163e-01 (3.54e-03) -	1.4225e-01 (6.79e-03) -	1.4056e-01 (9.01e-03) -	1.7657e-01 (1.76e-02) -	1.7905e-01 (9.86e-03) -	2.2619e-01 (4.92e-03) -	**2.5722e-01 (3.97e-03)**
P-n60-k15	1.7552e-01 (1.16e-02) -	1.5031e-01 (1.07e-02) -	1.4371e-01 (1.17e-02) -	2.1721e-01 (1.24e-02) -	1.7736e-01 (1.33e-02) -	2.4445e-01 (2.22e-02) -	**2.9742e-01 (6.38e-03)**
P-n65-k10	1.5738e-01 (5.62e-03) -	2.1433e-01 (3.22e-03) -	2.1241e-01 (2.50e-03) -	2.1344e-01 (5.30e-03) -	2.2059e-01 (8.07e-03) -	2.4478e-01 (3.57e-03) -	**2.6924e-01 (9.50e-03)**
P-n70-k10	1.8201e-01 (2.99e-03) -	2.3766e-01 (7.56e-03) -	2.3747e-01 (7.78e-03) -	2.6297e-01 (1.22e-02) -	2.6791e-01 (8.42e-03) -	3.0945e-01 (1.16e-02) -	**3.2810e-01 (9.14e-03)**
P-n76-k4	6.0259e-02 (7.31e-18) -	2.3958e-01 (2.48e-03) -	2.3902e-01 (1.90e-03) -	2.4275e-01 (7.36e-03) -	2.5789e-01 (4.77e-03) -	2.7248e-01 (6.31e-03) -	**2.9345e-01 (3.70e-03)**
P-n76-k5	1.3310e-01 (2.93e-17) -	2.0831e-01 (2.19e-03) -	2.0625e-01 (6.52e-04) -	2.3431e-01 (1.27e-02) -	2.3475e-01 (3.73e-03) -	2.6437e-01 (3.77e-03) -	**2.7355e-01 (4.97e-03)**
P-n101-k4	1.7525e-01 (5.81e-04) -	2.0554e-01 (9.40e-04) -	2.0493e-01 (3.21e-04) -	2.4779e-01 (1.96e-02) -	2.6438e-01 (7.12e-03) -	2.7998e-01 (1.19e-02) -	**3.2611e-01 (2.00e-03)**
+/-/=	0/24/0	0/24/0	0/24/0	0/24/0	0/24/0	0/24/0	

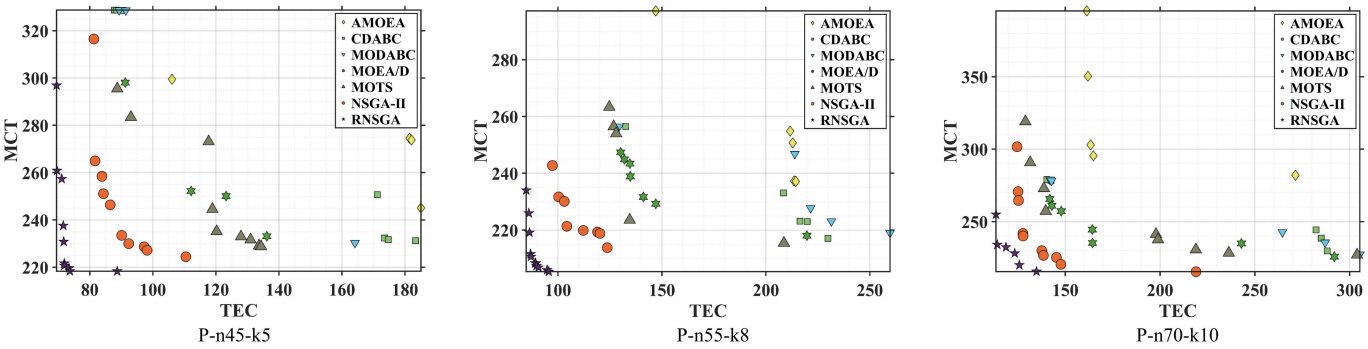


Fig. 8. Distribution of PF for representative test instances.

remaining unoptimized for extended periods. The *WNL* archive mechanism introduced by MODABC represents a valuable innovation, attempting to adjust interaction operator usage through accumulated problem-solving experience. However, this mechanism requires longer learning cycles to function effectively when confronting large-scale complex problems. This also indicates its limited capability in solving problems within a constrained time budget.

CDABC builds on MODABC by adding a targeted optimization process for the most time-consuming robot, which helps improve the solution quality. However, it fails to completely resolve the fundamental deficiencies inherent in MODABC. It randomly selects a robot's *TS* from the solution and attempts to optimize it with the *TS* of *MTC* using neighborhood operators. While this alleviates the workload of the heaviest robot, it does not adequately address the imbalance in robot working times.

AMOEA further combines problem-specific characteristics to enhance local search capabilities. It primarily focuses on the *TS*s of the *MTC* and *LTC*, facilitating a more balanced distribution of work times among robots. However, the random exchange of task points between two routes leads to low optimization efficiency. Moreover, it applies local search operators to multiple solutions in the population, consuming substantial computational resources and severely impacting the overall search process.

Due to the representation of solutions in these population-based optimization algorithms, their operators work on each

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TABLE III
AVERAGE RESULTS OF DEFAULT SOLUTIONS

Problems/ (TEC, MCT)	AMOEA	CDABC	MODABC	MOEA/D	MOTS	NSGA-II	RNSGA
P-n16-k8	(16,72)	(16,93)	(15,59)	(18,54)	(16,58)	(17,54)	**(14,53)**
P-n19-k2	(40,248)	(40,252)	(48,245)	(59,252)	(38,246)	**(43,244)**	**(30,248)**
P-n20-k2	(42,284)	(37,281)	(37,281)	(37,281)	(37,281)	(38,242)	**(32,242)**
P-n21-k2	(43,237)	(37,233)	(36,233)	(37,233)	(35,240)	(33,233)	**(31,232)**
P-n22-k2	(45,247)	(38,248)	(38,248)	(38,248)	**(37,242)**	(34,246)	**(32,246)**
P-n22-k8	(2037,6872)	(1983,8077)	(1900,8044)	(1877,7126)	(1933,6968)	(1810,5609)	**(1693,5261)**
P-n23-k8	(25,81)	(23,81)	(22,81)	(27,79)	(23,81)	(26,72)	**(19,64)**
P-n40-k5	(75,253)	(76,286)	(74,286)	(76,286)	(71,264)	(71,229)	**(60,214)**
P-n45-k5	(106,299)	(92,329)	(194,241)	(110,234)	(92,329)	**(109,231)**	**(70,233)**
P-n50-k10	(76,199)	(75,199)	(74,199)	(75,199)	(75,199)	**(70,199)**	**(66,210)**
P-n50-k7	(97,247)	(95,285)	(92,285)	(94,283)	(95,285)	(85,267)	**(77,228)**
P-n50-k8	(88,273)	(86,257)	(85,257)	**(146,210)**	(84,244)	(82,235)	**(72,223)**
P-n51-k10	(80,244)	(68,201)	(68,201)	(109,165)	(64,201)	(67,157)	**(60,154)**
P-n55-k10	(94,263)	(92,235)	(90,235)	(92,235)	(87,205)	**(96,178)**	**(76,225)**
P-n55-k15	(82,182)	(76,173)	(126,170)	(74,155)	(75,173)	(67,152)	**(65,123)**
P-n55-k7	(120,330)	(121,296)	(118,296)	(271,271)	(121,296)	(154,258)	**(90,312)**
P-n55-k8	(148,297)	(136,256)	(214,230)	(153,221)	(132,256)	(110,227)	**(88,212)**
P-n60-k10	(108,244)	(107,252)	(102,252)	(248,193)	(107,252)	(95,242)	**(87,183)**
P-n60-k15	**(94,148)**	(89,208)	(88,208)	(132,153)	(88,208)	(96,152)	**(77,157)**
P-n65-k10	(134,273)	(128,221)	(124,221)	(237,208)	(128,221)	**(113,221)**	**(102,248)**
P-n70-k10	(161,395)	(144,279)	(140,278)	(144,279)	(137,273)	(128,269)	**(114,249)**
P-n76-k4	(347,793)	(294,582)	(263,582)	(317,571)	(268,582)	**(256,559)**	**(223,572)**
P-n76-k5	(265,565)	(225,488)	(223,488)	(225,488)	(220,535)	(214,488)	**(186,486)**
P-n101-k4	(391,817)	(369,721)	(357,720)	(407,669)	(337,721)	**(345,651)**	**(278,663)**
Non-domination	1	0	0	1	1	7	**24**

robot's *TS* within each solution. In contrast, MOTS and RNSGA leverage single-point search to optimize routes individually. This allows for finer adjustments to solutions. Ultimately, the optimized routes are aggregated into multiple *TS*s assigned to robots. MOTS employs a memory strategy to accumulate search experience and avoid local optima. However, the process of generating neighboring solutions still involves significant randomness. When the number of tasks increases, its global search capability decreases sharply, consuming substantial computational resources without significantly improving results.

RNSGA, on the other hand, uses RBM and PBOM to generate neighboring solutions in a targeted manner. RBM incrementally fills empty routes, gradually balancing the workload. ISA adjusts the route to prioritize transferring tasks with weak associations from *MTC* to *LTC*, thus reducing extra moving distances. PBOM further refines adjacent routes' tasks, preventing excessive dispersion within the same route and minimizing moving time.

Additionally, we implement the multi-problem Wilcoxon test to examine the correlation of the results, listed in Table I in Section S-V. The differences between RNSGA and each competitor are evident. In each row, the indicator R^+ is larger than R^-, indicating RNSGA's superiority over all comparative algorithms. To provide a comprehensive visualization of the algorithms' overall performance, Fig. 9 presents the rankings obtained through the Friedman statistical test [43]. The results demonstrate that RNSGA consistently achieves the top position across both performance metrics, indicating its superior optimization capability.

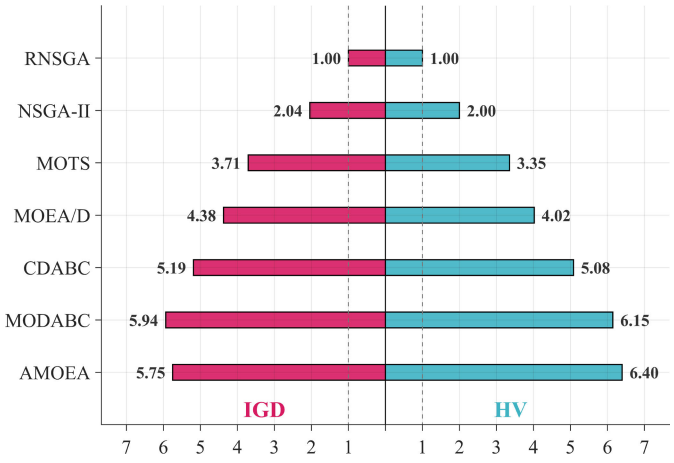


Fig. 9. Ranking of Friedman test results.

C. Analysis of the Default Solution and Case Visualization

Multi-objective optimization yields non-dominated solutions that minimize objective conflicts. These solutions provide decision-makers with optimal choices based on their preferences. When no specific preferences exist, this paper suggests a default solution. A higher HV value indicates a solution's objectives are further from the reference point (worst point). Unlike IGD, HV's reference point remains independent of comparative algorithms, ensuring unbiased evaluation. Therefore, this paper defaults to outputting the solution with the highest HV value.

Following the experimental setup in Section V-A, this section focuses on analyzing single default solutions. Table III presents the average (TEC, MCT) obtained from ten independent runs for each test instance. Non-dominated solutions achieved by the comparative algorithms are highlighted, with their total counts summarized in the last row. The results demonstrate that, consistent with analysis in previous subsections, RNSGA maintains a decisive advantage over other comparative algorithms. Although RNSGA's solutions are mutually non-dominated with other algorithms in some problem instances, the magnitude of its superiority in one objective significantly outweighs its minor disadvantages in the other. For example, in the P-n19-k2 problem, RNSGA accepts a mere 1.6% disadvantage in MCT compared to

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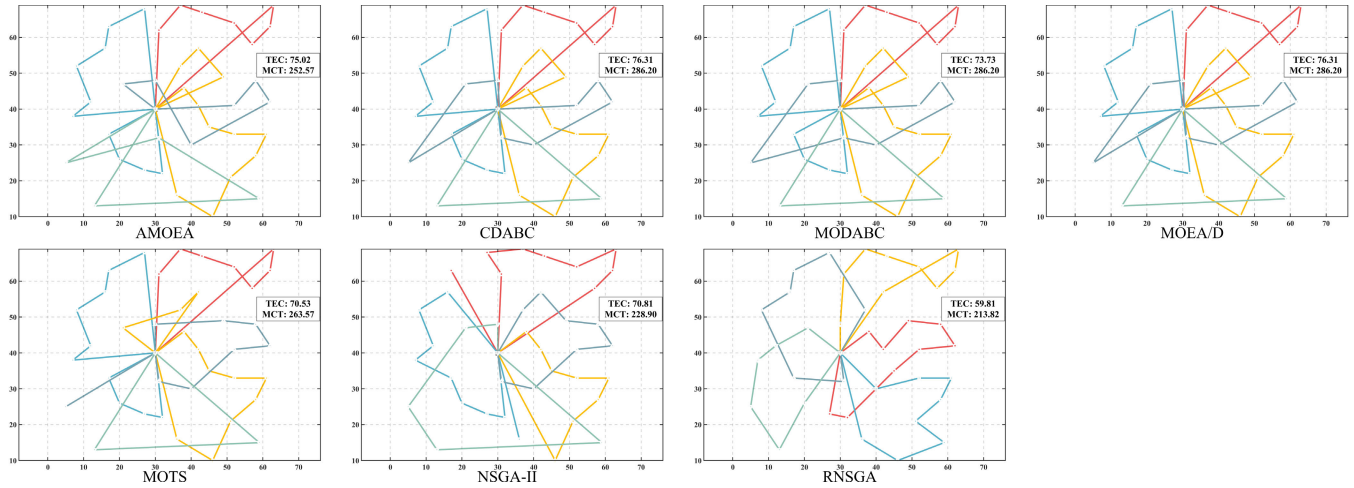


Fig. 10. Case visualization.

NSGA2 while achieving a substantial 30% improvement in TEC. Similarly, for problems P-n45-k5 and P-n50-k8, the objective value improvements correspond to approximately (35.7%, -0.8%) and (50.6%, -6.1%), respectively. This pattern reveals that RNSGA consistently delivers substantial energy efficiency improvements with only small time penalties, representing an excellent trade-off.

For intuitive demonstration, we select a test instance (P-n40-k5) to illustrate the paths of each vehicle. As shown in Fig. 10, different colors represent different vehicle routes. The routing results reveal that all algorithms exhibit similar task execution strategies: prioritizing tasks that are farther from the depot but with smaller demands, while arranging tasks that are closer to the depot with larger demands. The experimental results reveal that CDABC, MODABC, and MOEA/D converge to similar local optima in this problem instance. NSGA-II allocates separate routes for task points with larger demand, which significantly reduced the MCT. The superior performance of RNSGA can be attributed to its methodological approach of decomposing complete solutions into independent paths for individual optimization. This decomposition, coupled with diverse search mechanisms, enables RNSGA to construct highly efficient route structures.

D. Other Experiments

Due to the limit on the numbers of pages, some other experiments are provided in the supplementary materials. In Section S-III, we calibrate the main parameter, $Pnum$, of RNSGA. In addition, in Section S-IV, we verify several important components include the solution initialization method, ISA, RBM, PBOM, and PRAM.

VI. Conclusion and Future Works

This paper integrates variable neighborhood search theory with population-based approaches, proposing the RNSGA to address the MRRS problem in transportation scenarios. Extensive experiments and analysis demonstrate the effectiveness and superiority of RNSGA. The algorithm's problem-solving ability is influenced by several methods and mechanisms, including the solution initialization method, ISA, RBM, PBOM, and PRAM. Due to the correlation between the operator design and the specific scenario, RNSGA cannot be directly applied to other MRRS problems. Nonetheless, the algorithm's design process, innovative mechanisms, and problem-solving approach provide valuable insights for developing other MRRS algorithms.

Despite the promising results, several limitations of the proposed RNSGA need to be addressed in future research. These include applicability of the modeling in practice, adaptability in dynamic environments, and parameter tuning efficiency.

Hence, in future works:

- Problem-driven models and solutions that are closer to real-world scenarios by incorporating field data collection and terrain-specific constraints will be designed [47].
- Uncertainties and dynamic conditions, such as sudden malfunctions and unexpected increases in market demand will be taken into consideration through predictive maintenance models and adaptive control mechanisms [48].
- A user-friendly application interface to enhance operational accessibility and cross-environment adaptability will be developed, facilitating the deployment of our algorithm across diverse agricultural scenarios.

Supplementary Materials

To provide a more complete presentation of this paper, supplementary materials are included in the appendix, which contain:

- Section S-I: pseudo-codes of algorithms $2-7$;
- Section S-II: computational complexity;
- Section S-III: parameter validation experiment;
- Section S-IV: ablation experiment;
- Section S-V: tables and figures of some mechanisms and experimental results.

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